

6.2	Apply the knowledge that approximately $\frac{2}{3}$ of observations lie within $\mu \pm \sigma$, and equivalent results for 2σ and 3σ .	Students should learn that approximately 95% of observations lie within $\mu \pm 2\sigma$ and approximately 99.8% of observations lie within $\mu \pm 3\sigma$.
6.3	Determine probabilities and unknown parameters with a normal distribution.	Students may use calculator functions to obtain information for a normal distribution directly and are advised to do so.
6.4	Apply the normal distribution to model real-world situations.	
6.5	Use the fact that the distribution of $\bar{X}$ has a normal distribution if X has a normal distribution.	Students are expected to know that $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$
6.6	Use the fact that the normal distribution can be used to approximate a binomial distribution under particular circumstances.	Students should learn that, as an alternative to using the exact binomial distribution, the normal distribution may be used to approximate a binomial distribution when: • $n \geq 20$ and $p \approx 0.5$ • $np > 10$ or $n(1-p) > 10$ Students will be expected to use a continuity correction when finding a probability if a normal approximation to a binomial distribution is used.

7 Correlation and linear regression

Subject content	Additional information
7.1 Calculate (only using appropriate technology – calculator) and interpret association using Spearman’s rank correlation coefficient or Pearson’s product moment correlation coefficient.	When evaluating Spearman’s coefficient, candidates will be expected to: • rank both variables consistently • rank tied values appropriately • use a calculator to obtain the value of Spearman’s rank correlation coefficient Students will be expected to find Pearson’s product moment correlation coefficients directly from the calculator. Interpretation of correlation coefficients should always be in the context of the question.
7.2 Use tables to test for significance of a correlation coefficient.	A test on a Pearson’s correlation coefficient requires the assumption that the population has a bivariate normal distribution.
7.3 Know the appropriate conditions for the use of each of these methods of calculating correlation and determine an appropriate approach to assessing correlation in context.	Pearson’s product moment correlation coefficient measures the strength of the linear relationship between 2 variables. For Spearman’s rank correlation coefficient, no assumptions are necessary regarding the distribution used.
7.4 Calculate (only using appropriate technology – calculator) and interpret the coefficients for a least squares regression line in context; interpolation and extrapolation, and use of residuals to evaluate the model and identify outliers.	Students will be expected to find the coefficients for a least squares regression line directly from the calculator. Students will be expected to know that $\text{residual} = y_i - a - bx_i$ for a least squares regression line of the form $y = a + bx$ Students will be expected to make a numeric evaluation of a residual when asked for the value of a specific residual(s) but may be asked to make general comments regarding residuals visually from a scatter graph with a line of best fit drawn on it.